

Customer Churn Analysis Report

Project Title

Reducing Telecom Customer Churn Through Behavioral and Revenue Analytics

1. Executive Summary

Customer churn remains one of the biggest challenges for subscription-based businesses, particularly in the telecommunications industry where competition is intense and switching barriers are low.

This project analyzes customer churn behavior using customer demographics, contract details, service usage patterns, support interactions, and revenue analytics to identify the major drivers of churn and provide actionable business recommendations.

The final analysis revealed that customer churn is strongly influenced by:

- Month-to-month contract structures
- Poor customer service experiences
- Competitive offers from rival telecom providers
- Short customer tenure
- High-risk low-value customer segments
- Weak customer onboarding among new accounts

The analysis also identified several retention opportunities:

- Long-term contracts significantly improve retention
- Group contracts reduce churn substantially
- High-value customers are more loyal
- Customer service calls serve as strong churn warning signals
- High-risk customer groups can be identified early using behavioral analytics

The final solution was delivered using Microsoft Power BI through three business-focused dashboards:

1. Executive Overview
2. Customer Behavior
3. Revenue Insights

These dashboards were designed to help executives understand:

- What is happening
- Why it is happening
- What the financial impact is
- What actions the business should take

2. Business Problem

Customer churn negatively affects:

- Revenue growth
- Customer lifetime value
- Profitability
- Operational efficiency
- Long-term business sustainability

The company needed a data-driven approach to:

- Understand the severity of the churn
- Understand why customers churn
- Identify high-risk customers
- Quantify revenue loss
- Improve customer retention strategies
- Prioritize retention investments

3. Business Objectives

The main objectives of this analysis were to:

1. Measure customer churn rate
2. Identify major churn drivers
3. Analyze customer behavior patterns
4. Understand revenue impact of churn
5. Segment customers by value and risk
6. Identify retention opportunities
7. Provide strategic business recommendations

4. Dataset Overview

The dataset contains telecom customer information including:

- Customer demographics
- Contract details
- Service usage behavior
- International activity
- Customer service interactions
- Charge metrics
- Churn status and churn reasons

Dataset Highlights

Metric	Value
Total Customers	6,677

Churned Customers	1,793
Retained Customers	4,884
Overall Churn Rate	26.85%
Total Revenue	\$7.24M
Revenue Lost to Churn	\$1.36M

5. Tools Used

Tool	Purpose
Microsoft Power BI	Data modeling, visualization, dashboarding
Power Query	Data cleaning and transformation
DAX	KPI calculations and customer segmentation

6. Data Preparation

The dataset was cleaned and transformed using Power Query.

Data Cleaning Steps

- Corrected data types
- Removed duplicates
- Handled missing values
- Standardized categorical values
- Created segmentation columns
- Built customer risk and value categories

7. Data Modeling

A star-schema-inspired model was used to improve organization and scalability.

Tables Created

Fact Table

- Datablel - Data

Dimension Tables

- DimCustomer
- DimContract
- DimGeography
- DimChurn
- DimSegmentation

This structure improved:

- Performance
- Maintainability
- DAX readability
- Dashboard development workflow

8. Customer Segmentation Framework

Customers were segmented using:

Customer Value Segmentation

Based on:

- Total Charges
- Monthly Charges
- Account Length

Segments:

- High Value
- Mid Value
- Low Value

Customer Risk Segmentation

Based on:

- Customer service calls
- Contract type
- Extra charges
- Account Length

Segments:

- High Risk
- Moderate Risk
- Low Risk

Combined Risk-Value Matrix

Customers were grouped into strategic categories such as:

- High Value – High Risk

- High Value – Low Risk
- Low Value – High Risk
- Mid Value – Moderate Risk

This framework enabled targeted retention strategies.

9. Dashboard Overview

The project was condensed into three storytelling dashboards.

Dashboard 1 — Executive Overview

Provides executives with a high-level understanding of:

- Churn severity
- Major churn drivers
- Geographic trends
- Customer demographics

Key Visuals

- KPI Cards
- Churn by Contract Type
- Churn by State
- Churn by Age Group
- Churn by Category
- Top Churn Reasons

Dashboard 2 — Customer Behavior

Analyzes customer behavior and identifies patterns associated with churn.

Key Visuals

- Customer Service Calls vs Churn
- International Usage vs Churn
- Churn Rate vs Account Length
- Data Usage Distribution
- Group Contracts vs Churn

Dashboard 3 — Revenue Insights

Quantifies the financial impact of churn and identifies revenue protection opportunities.

Key Visuals

- Revenue by Contract Type
- Revenue Lost by Churn Category

- Revenue by State
- Customer Value Segmentation
- Revenue at Risk
- Risk-Value Segmentation Matrix

10. Key Findings & Insights

1. Churn Rate Is Critically High

The overall churn rate of 26.85% indicates major retention issues within the business.

This suggests:

- Weak customer loyalty
- Competitive vulnerability
- Dissatisfaction among key customer segments

2. Month-to-Month Contracts Are the Biggest Churn Driver

Contract Type	Churn Rate
Month-to-Month	46.27%
One Year	11.31%
Two Year	2.79%

Month-to-month customers are significantly more likely to churn compared to long-term contract customers.

At the same time, two-year contracts generate the highest total revenue at approximately \$3.05M, accounting for nearly 42% of total company revenue.

This demonstrates that long-term customers are both:

- More loyal
- More profitable

3. Competitors Are the Largest External Threat

The leading churn reasons include:

- Competitor made better offer
- Competitor had better devices
- Competitor offered more data

This indicates strong competitive pressure in the telecom market.

Competitor-related churn dominates both customer loss and revenue loss.

4. Customer Service Strongly Influences Churn

Customers with repeated support interactions churn at extremely high rates.

Key findings:

- Customers with 5 service calls show 100% churn
- Churned customers average 2.4 support calls
- Retained customers average only 0.37 calls

Customer support quality is therefore a major churn driver.

This makes support interactions one of the strongest predictive indicators of future churn.

5. Longer Customer Tenure Reduces Churn

Customers with longer account tenure demonstrate significantly lower churn rates.

This suggests that:

- Early-stage customers are most vulnerable
- Onboarding quality matters heavily
- Retention strategies should focus strongly on new customers

6. International Usage Reveals a High-Risk Customer Group

Customers using between 1–50 international minutes experience an extremely high churn rate of 64.42%.

These customers also show:

- Very short account lengths
- Low customer maturity
- Early dissatisfaction behavior

This indicates that new international users represent a critical churn-risk segment.

7. Group Contracts Improve Retention

Grouped contracts reduce churn by more than 20%.

This indicates:

- Customers in groups are more loyal
- Family/business plans improve stickiness
- Multi-line subscriptions increase switching barriers

8. Most Customers Are Low Data Users

More than 75% of customers use less than 10GB of data monthly.

This suggests:

- Many customers may be overpaying for services
- Some plans may not align with customer usage patterns
- Perceived pricing fairness could influence retention

9. High-Value Customers Are More Loyal

Churn drops significantly as customer value increases.

Segment	Churn Rate
Low Value	~31%
Mid Value	~19%
High Value	~16%

This suggests that high-value customers perceive greater value from the service and demonstrate stronger loyalty.

10. High-Risk Segments Dominate Churn

Customer segmentation analysis shows:

- High-risk customers dominate overall churn
- Low-value customers are the most volatile segment
- High-value customer retention remains relatively strong
- Mid-value customers represent major growth opportunities

This segmentation framework helps prioritize retention investments strategically.

11. Churn Creates Significant Financial Impact

Metric	Value
Total Revenue	\$7.24M
Revenue Lost	\$1.36M
Revenue at Risk	\$151K

Revenue loss is heavily concentrated along the East Coast and major urban corridors. This indicates that churn is not only a customer issue but also a major revenue leakage problem.

11. Business Recommendations

Recommendation 1 — Promote Long-Term Contracts

The business should aggressively encourage migration from month-to-month plans toward:

- One-year contracts
- Two-year contracts

Possible incentives:

- Discounts

- Device upgrades
- Loyalty rewards
- Bundled packages

Recommendation 2 — Improve Customer Service Quality

Because support experience strongly influences churn, the company should:

- Retrain support staff
- Improve response times
- Implement escalation monitoring
- Improve complaint resolution processes

Recommendation 3 — Build an Early Churn Warning System

Customers with:

- Multiple service calls
- Short tenure
- Extra charges
- Month-to-month contracts

should automatically be flagged as high-risk customers.

Recommendation 4 — Protect High-Value Customers

The company should prioritize retaining premium customers using:

- VIP support
- Loyalty programs
- Personalized offers
- Proactive engagement

Recommendation 5 — Expand Group Plans

The business should promote:

- Family bundles
- Multi-line subscriptions
- Business group contracts

because grouped customers show significantly better retention.

Recommendation 6 — Improve Competitive Positioning

To reduce competitor-driven churn, the company should:

- Benchmark competitor pricing
- Improve device offerings
- Introduce better value bundles
- Improve pricing transparency

12. Business Impact

If the business successfully reduces churn, potential benefits include:

- Increased revenue retention
- Higher customer lifetime value
- Reduced acquisition costs
- Improved customer loyalty
- Better profitability

Even a small reduction in churn could preserve hundreds of thousands of dollars in revenue.

13. Conclusion

This analysis demonstrates that customer churn is heavily influenced by:

- Contract structure
- Customer experience
- Competitive pressure
- Customer tenure
- Revenue/value segmentation

The findings show that retention strategies should focus on:

- Long-term contracts
- Customer support improvement
- Early churn detection
- High-value customer protection

The final Power BI solution provides an executive-level framework for understanding churn, prioritizing retention efforts, and reducing revenue leakage.